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Project Background

UCSD Health's CPM team oversees a multi-billion dollar portfolio of construction projects. Without predictive tools, PMs rely on intuition when forecasting timelines and cash flow to stakeholders.

Working with Executive Director Chris Hickman, we turned 14 years of eBuilder spending data into reusable forecasting models – accessible via a web tool.

228

Projects

\$6.6B

Portfolio

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Aims & Objectives

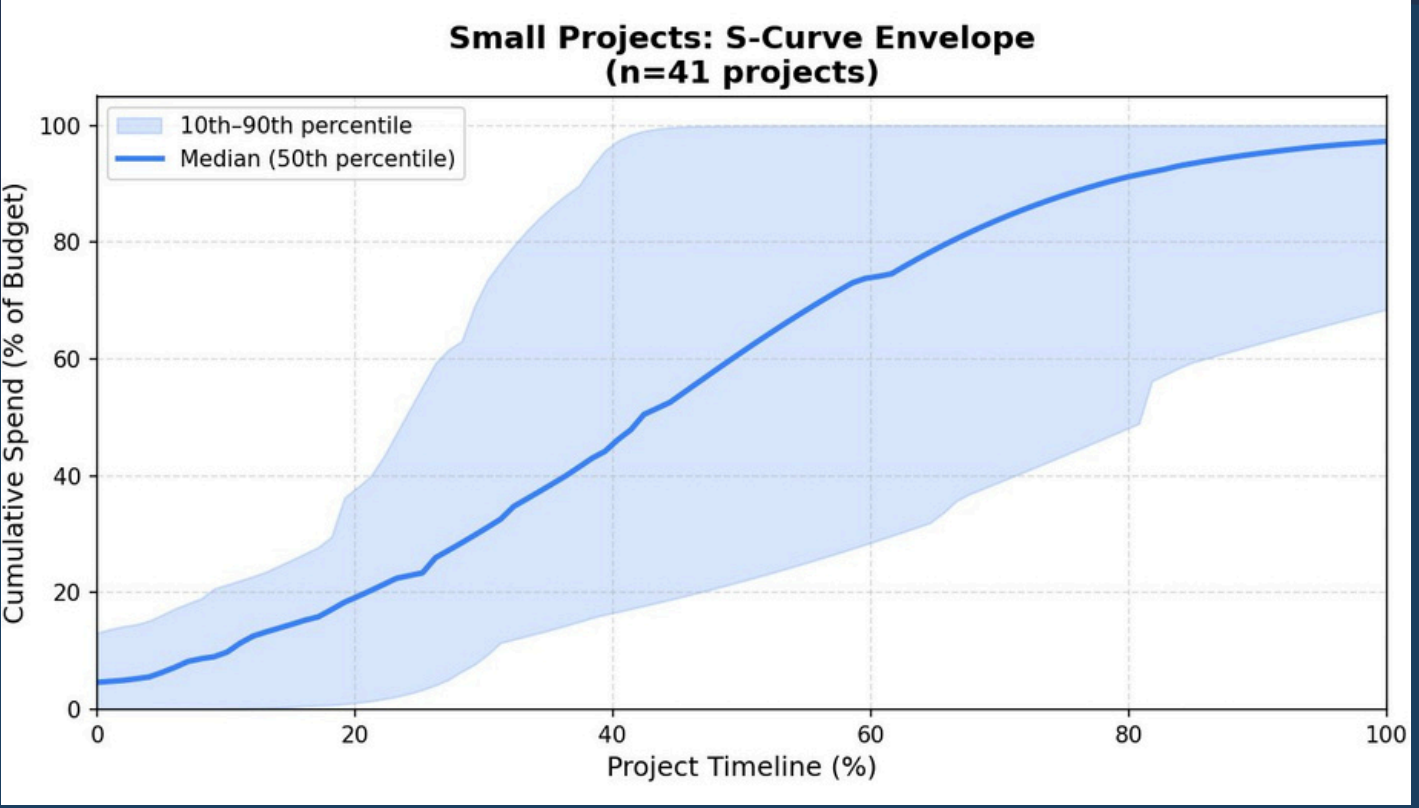
- Develop S-curve cash flow models by project size tier
- Classify portfolio: Small / Medium / Large / Mega
- Quantify CPM overhead (Line 6) as % of total cost
- Build PM-facing web tool: budget + sq ft → predicted spend curve

0 4 — Core Deliverable

S-Curve Cash Flow Envelopes by Tier

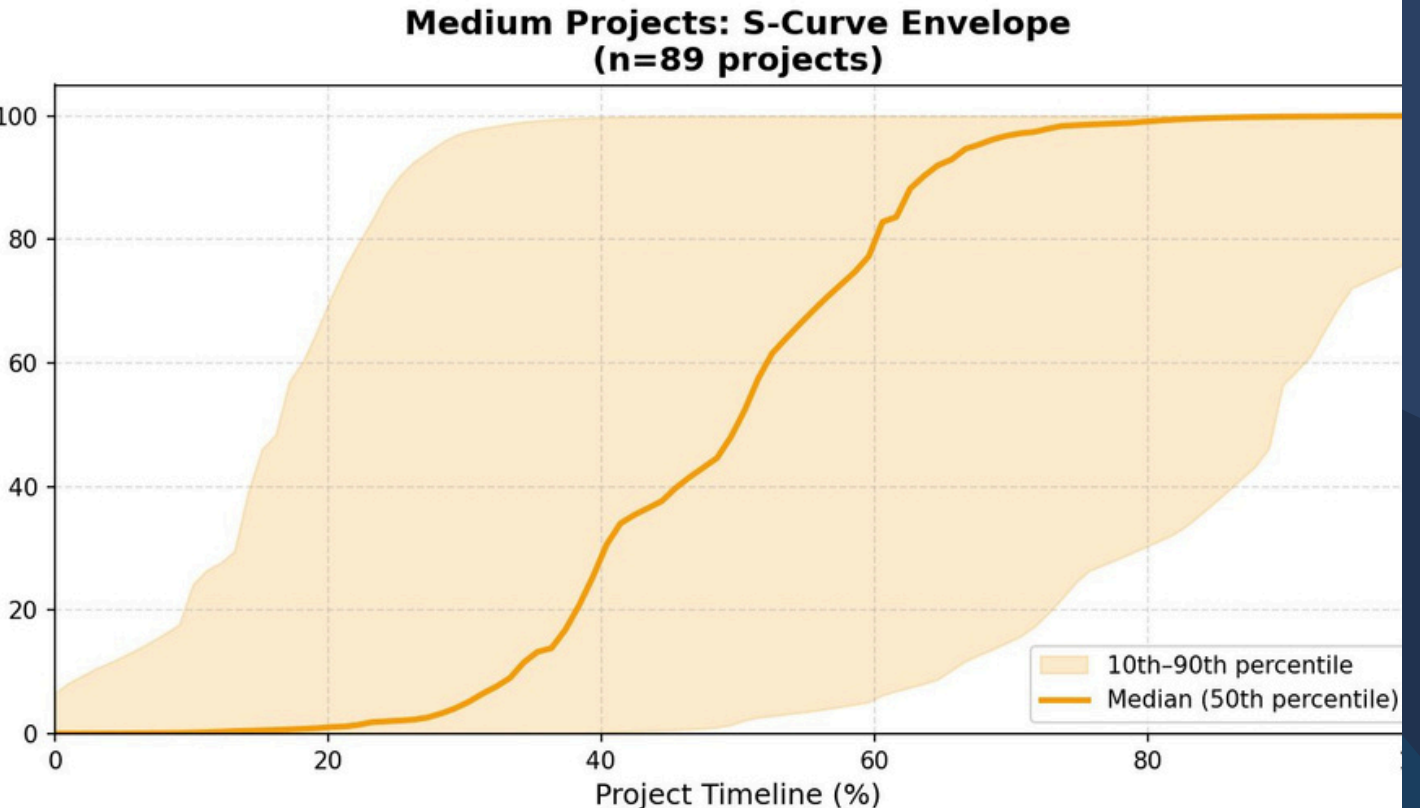
Logistic model fit · Median + 10th–90th pct band · Total R²s: ~0.62, ~0.76

Small Projects: S-Curve Envelope (n=41 projects)



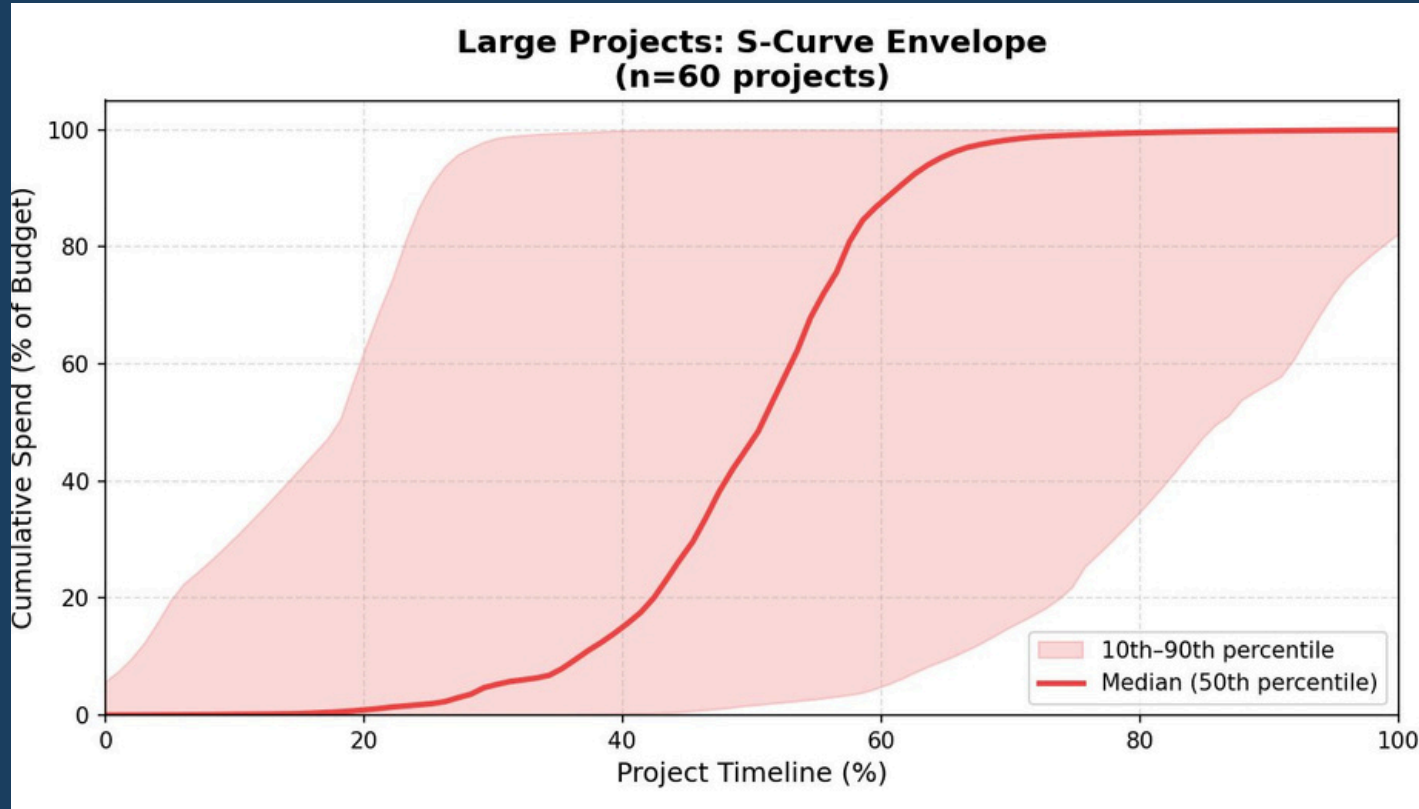
Small <\$250K · n=41 · med 18 mo

Medium Projects: S-Curve Envelope (n=89 projects)



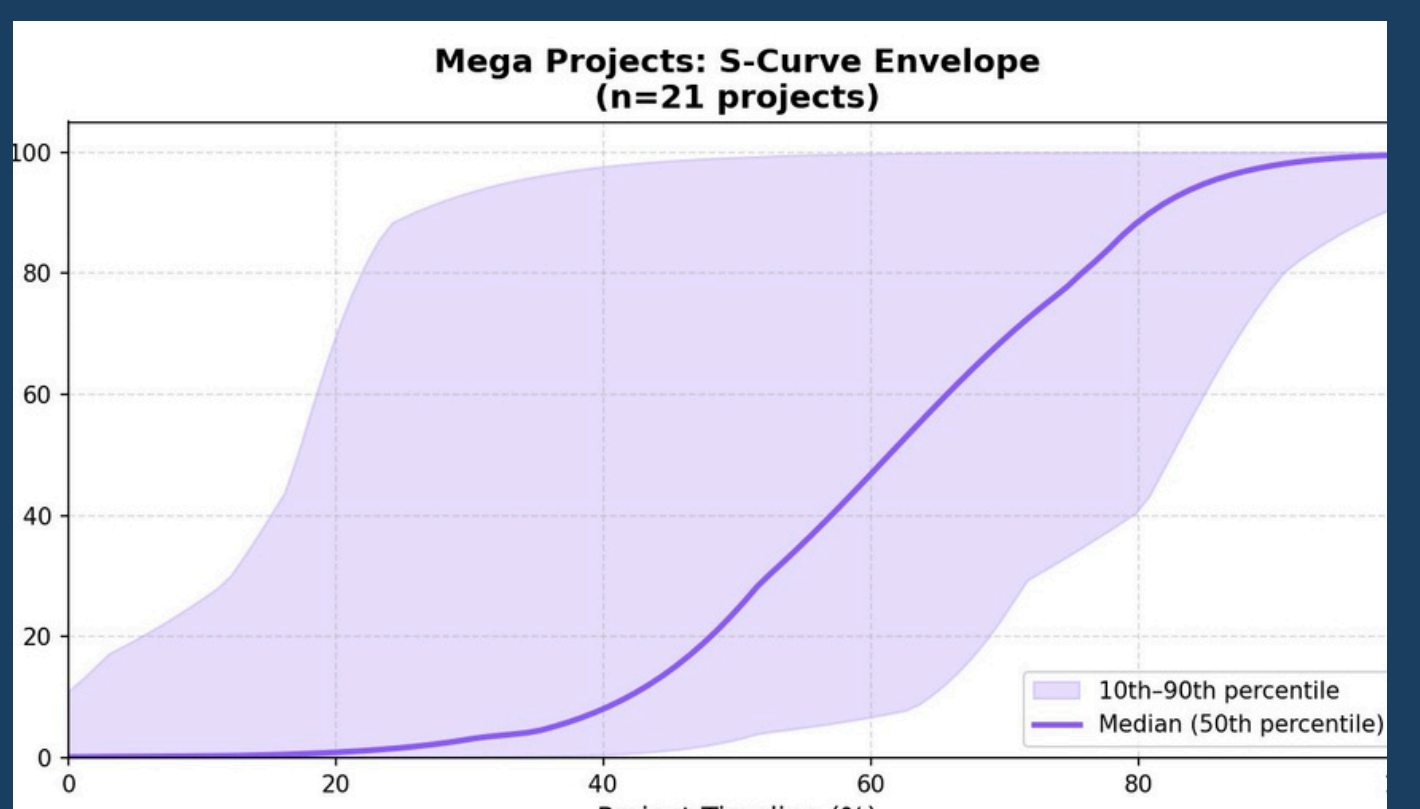
Medium \$250K–\$5M · n=89 · med 44 mo

Large Projects: S-Curve Envelope (n=60 projects)



Large \$5M–\$50M · n=60 · med 54.5 mo

Mega Projects: S-Curve Envelope (n=21 projects)



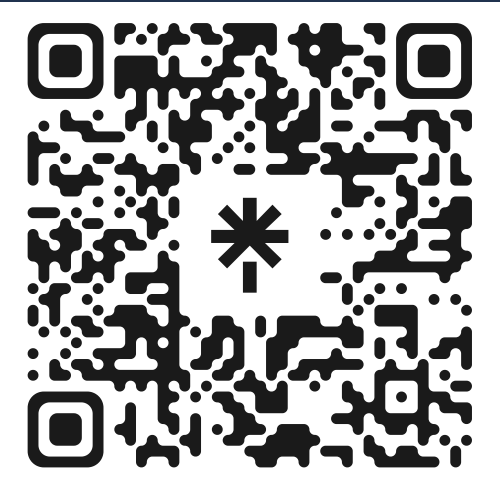
Mega >\$50M · n=21 · med 67 mo

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Outcomes & Deliverables

- Built an interactive tool that predicts project timelines and cost curves
- Trained on 14+ years of data from 200+ projects (\$6.6B+)
- Generates project-specific forecasts based on size and budget
- Visualizes how spending evolves over time through dynamic cost curves

Virtual Tutorial :



Capital Project Cashflow Predictor

Input Method

☒ CSV Upload

☐ Manual Entry

Curve Type

☐ Total

☒ PM Fees and Overhead

Upload Project CSV

5171 Hillcrest Outpatient Pavilion and Parking (965230) 12.4KB

+

Gross Sq Footage

251000.00

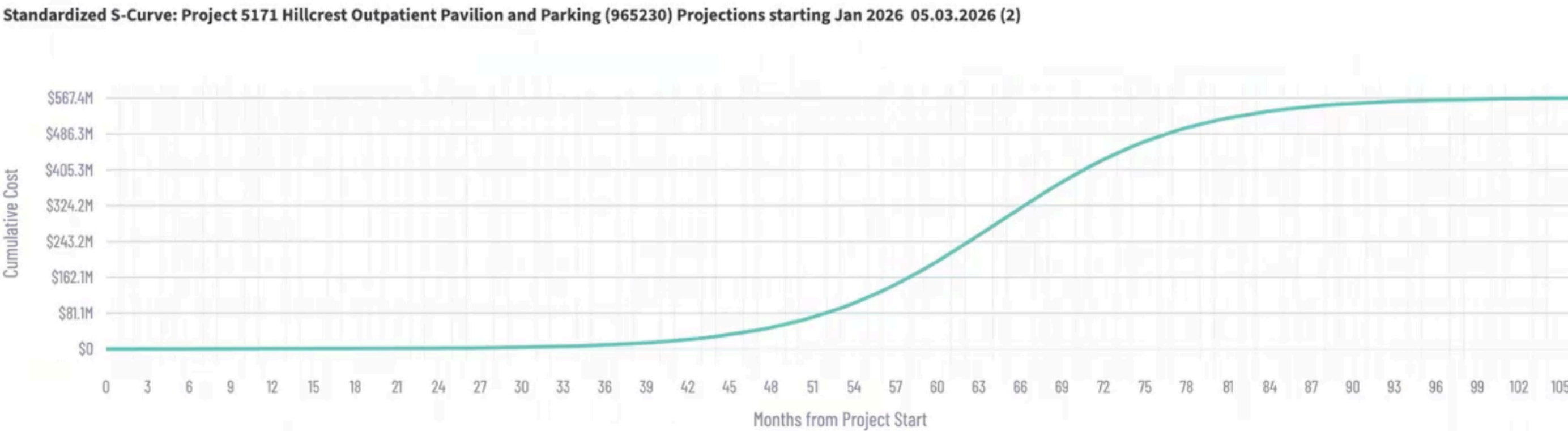
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Generate S-Curve

Capital Project Cashflow Predictor

Standardized S-Curve: Project 5171 Hillcrest Outpatient Pavilion and Parking (965230) Projections starting Jan 2026 05.03.2026 (2)



Duration

107.3 months

Growth Rate (k)

0.1453

Midpoint (t₀)

64.3 months

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Project Journey & Milestones

Dec 2025

Discovery & Kickoff

Met with Chris Hickman to define scope. Mapped data sources, aligned on deliverables: predict project duration and cash flow curves from 14 years of internal eBuilder data.

Jan 2026

Data Intake & Cleaning

Parsed eBuilder spreadsheets. Identified 869 complete projects. Isolated 228 with gross sq ft data for model training. Standardized IDs and monthly spend columns.

Feb 2026

Baseline EDA & Tier Classification

Formalized Sm/Med/Lg/Mega tiers. Median duration ranges from 18 months (Small) to 67 months (Mega).

Mar 2026

S-Curve Model Development

Fit logistic S-curves per project. Extracted k (growth rate) and t₀ (spending midpoint). Total R²s: ~0.62, ~0.76

Apr 2026

Line 6 Overhead Analysis

CPM internal costs as % of total spend. Key finding: overhead drops from 39% (Small) to 1% (Mega).

May 2026

Capital Project Cashflow Predictor

Integrated sq ft as primary feature. RandomForest and LightGBM, model ensemble. Deployed Capital Project Cashflow Predictor.

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Learning Outcomes

- Applied ensemble ML to real institutional data
- Navigated 14yr multi-format eBuilder data
- Stakeholder communication w/ exec leadership
- Shipped a production tool used by CPM team



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Next Steps

- UC System Expansion — Scale to all UC campuses
- Higher Level Analysis - Integrated with NLP and Descriptions

Quick Estimate

New Construction · 50,000 sq ft · La Jolla, CA 2025

Early-stage ballpark only — the UCSD model has ±82% error at this input fidelity. Triangulate all three sources. For formal estimates, use the S-Curve Generator with eBuilder line-item data.

UCSD HISTORICAL

64 similar projects

Total EAC

\$34.39M

Line 6 (PPM Fee)

\$470K

Predicted Duration

57 mo

P25: \$3.98M Median: \$9.59M P75: \$45.18M

Trained on 238 UCSD eBuilder projects · ±82% MAPE

CA REGIONAL BENCHMARK

LOW

\$22.50M

\$450/sqft

MID

\$35.00M

\$700/sqft

HIGH

\$55.00M

\$1100/sqft

Duration Range

30–84 mo

DGS CCCI + ENR LA + San Diego coastal adjustment (2024–2025)

AI MARKET ESTIMATE

Llama 3.3 70B (Groq)

LOW

\$35.00M

MID

\$40.00M

HIGH

\$45.00M

Duration Range

24–36 mo

"Estimates are based on historical data and industry benchmarks for institutional projects in San Diego, considering the unique requirements and complexities of the UC San Diego campus."

La Jolla, CA context - Groq free tier